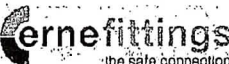


(A01) ERNE Fittings GmbH Hauptstraße 48 6824 Schlin Austria  ernefittings the safe connection	 ernefittings the safe connection	(A02) INSPECTION CERTIFICATE 3.1 EN10204:2004 (A03) No.: 2021-010603-SO Rev.: 0 Date: 2021-04-07 Page: 1 / 4
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(A06) Customer Erne Fittings USA, Inc. 9725 Kirkton TX 77095 Houston United States	Consignee Dodson Global - Houston 2 2750 Holmes Road TX 77051 Houston United States
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Order (A07, A08)	
Erne order no.	000796CP
Erne order pos.	0330
Customer order no.	272909-00
Order date	2020-10-30

Requirements of the order (B01, B02, B03, B04)
cap MSS SP-75-14" + CSA Z245.11-17 material : WPHY 60/WPHY 65/Gr. 448 Cat-II TR-fitting : MSS SP-75-2014 : CSA Z245.11-17 TR-addit. fitting : NACE MR0175/ISO15156-2:2015A.2 : NACE MR0103/ISO17945:2015 : 13.1 & 15 : For WPHY65: ASTM A860-18 certificate : EN 10204:2004/3.1 rust protection: Millstandard

Item specification (B01)	
Manufacturer item no.	178852
Product form	cap
Article designation	CAP-S-WPHY60/65/448CatII-16"-XS
Steel grade	WPHY_60/WPHY_65/Gr_448_Cat-II

Product dimensions (nominal) (B09, B10, B11)	
Diameter	406.40 mm
Wall thickness	12.70 mm

Product (B04, B07, B08)	
Code No.	HCT2N
Quantity	10 pcs
Manufacturing process	Hot formed

Marking, Identification
 AUSTRIA Z245.11-17 448 M46C SS A860 WPHY60/WPHY65 QT HCT2N 16" XS 

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Pre material

(C70)

Heat no.	Starting material
1TAT58	plate

Heat analysis

(C71)

Heat no.	C	Si	Mn	P	S	Cr	Mo	Ni	Cu	V
1TAT58	0.160 %	0.20 %	1.45 %	0.016 %	0.003 %	0.04 %	< 0.01 %	0.02 %	0.02 %	< 0.01 %

Heat no.	Nb	Al	B	Ti	N	CE _{IIW}
1TAT58	< 0.01 %	0.033 %	< 0.0003 %	< 0.01 %	0.01 %	< 0.41

Product analysis

(C81)

Heat no.	C	Si	Mn	P	S	Cr	Mo	Ni	Cu	V
1TAT58	0.154 %	0.24 %	1.45 %	0.008 %	0.001 %	0.02 %	0.01 %	0.03 %	0.03 %	0.00 %

Heat no.	Nb	Al	B	Ti	N	CE _{IIW}	CE _N
1TAT58	0.02 %	0.030 %	< 0.0001 %	0.00 %	0.005 %	0.40	< 0.39

Formula legend

(C71)

Code	Name	Formula
CE _{IIW}	Carbon Equivalent (IIW)	$C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$
CE _N	Carbon Equivalent (N)	$C + (0.75 + 0.25 \cdot \tanh(20 \cdot (C - 0.12))) \cdot (Si/24 + Mn/6 + Cu/15 + Ni/20 + (Cr+Mo+Nb+V)/5 + 5B)$

Heat treatment

(B04)

Process	Temperature	Cooling
Quenched and tempered	920 °C	Cooling in water
	640 °C	Cooling in still air

Hardness test

(C30, C31)

Test report no.	Method	Min	Max
14293	HB	-	184

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Tensile test

(C01, C02, C03, C10, C11, C12, C13)

Sample no.	Sample orientation	Sample size	Temperature	Yield Strength	Tensile strength	Elongation
				R _{p0.2}		
MXP_455	Transversal	Round specimen Ø 8.91 mm L: 35.60 mm 4D	+22 °C	71500 PSI	88000 PSI	29 %

Notched bar impact test

(C40, C41, C42, C43, C44)

Sample no.	Sample orientation	Temperature	Dimension of test sample			Work	Lat. width	Shear fracture
			bo	So	Notch shape			
MXP_455	Longitudinal	-46 °C	10.0 mm	80.0 mm ²	V-Notch	237.0 J 260.0 J 251.0 J Ø 249.3 J	2.44 mm 2.51 mm 2.49 mm	
MXP_456	Transversal	-46 °C	10.0 mm	80.0 mm ²	V-Notch	139.0 J 189.0 J 130.0 J Ø 152.7 J		75.0 % 75.0 % 75.0 %

Additional tests

(D02, D51)

Test name	Test report no.	Test standard	Conditions	Rate	Result
Visual inspection and dimensional check					Without objection

Statement of Compliance

(Z01)

The products were manufactured, sampled, tested and inspected in accordance with the specifications and were found to meet the requirements.

Inspection representative

(A05, Z02, Z03)

Date	2021-04-07
Validated by	Inspection Representative
Name / Signature / Stamp	Sommer Stefanie
Telephone	+43 5524 501 328
Mail	stefanie.sommer@ernefittings.com

The certificate is issued by the computerized system and is valid without signature.

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Indication in parentheses correspond to attributes according to EN 10168

Official approvals (Z05)		
Approved by	Name	Certificate No.
TÜV SÜD Industrie Service GmbH	AD 2000 - Merkblatt W0; Pressure Equipment Directive (PED) 2014/68/EU Annex I, Paragraph 4.3 ; EN 764-5, Para. 4.2	DGR - 0036 - QS - W 129 / 2002 / MUC

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In order to compare this document with the original document you have to scan the QR code. If you have doubts about the authenticity do not hesitate to contact us by email: check.certificates@ernefittings.com

